



U.S. Department
of Transportation

**Pipeline and Hazardous
Materials Safety
Administration**

February 14, 2022

1200 New Jersey Avenue, SE
Washington, DC 20590

DOT-SP 10370
(FOURTEENTH REVISION)

EXPIRATION DATE: 2026-01-31

(FOR RENEWAL, SEE 49 CFR 107.109)

1. GRANTEE: Welker, Inc.
Sugar Land, TX
2. PURPOSE AND LIMITATIONS:
 - a. This special permit authorizes the manufacture, mark, sale, and use of non-DOT specification aluminum cylinders conforming with all regulations applicable to a DOT Specification 3E cylinder, except as specified herein, for the transportation in commerce of the hazardous materials authorized by this special permit. This special permit provides no relief from the Hazardous Materials Regulations (HMR) or the International Civil Aviation Organization's Technical Instructions for the Safe Transport of Dangerous Goods by Air (ICAO TI) other than as specifically stated herein. The most recent revision supersedes all previous revisions.
 - b. The safety analyses performed in development of this special permit only considered the hazards and risks associated with transportation in commerce.
 - c. This special permit serves as an "Approval" under Part 6, Chapter 5, Paragraph 5.1.1.2 of the International Civil Aviation Organization's Technical Instructions (ICAO TI), and as a "Competent Authority Approval" as defined under 49 CFR § 107.1.
 - d. In accordance with 49 CFR 107.107(a), party status may not be granted to a manufacturing special permit. These packagings may be used in accordance with 49 CFR 173.22a.
3. REGULATORY SYSTEM AFFECTED: 49 CFR Parts 106, 107 and 171-180 and the ICAO TI.

4. REGULATIONS FROM WHICH EXEMPTED: 49 CFR §§ 173.201, 173.202, 173.203, 173.302a(a)(1), and 173.304(a) in that non-DOT specification cylinders are not authorized, except as specified herein; and § 173.301(f)(2) and § 177.840(a)(1) in that the pressure relief device on the product side of the cylinder is in contact with liquid product at all times.
5. BASIS: This special permit is based on the application of Welker, Inc., dated January 27, 2022, submitted in accordance with § 107.109.
6. HAZARDOUS MATERIALS (49 CFR 172.101):

Hazardous Material Description			
Proper Shipping Name	Hazard Class/ Division	Identification Number	Packing Group
Helium, compressed	2.2	UN1046	N/A
Hydrocarbon gas mixture, liquefied, n.o.s.	2.1	UN1965	N/A
Methane, compressed or Natural gas, compressed	2.1	UN1971	N/A
Nitrogen, compressed	2.2	UN1066	N/A
Petroleum crude oil	3	UN1267	I, II, or III
Petroleum gases, liquefied	2.1	UN1075	N/A

7. SAFETY CONTROL MEASURES:

- a. PACKAGING: Prescribed packaging is an aluminum non-DOT specification cylinder having a marked service pressure of 600 psi, constructed from seamless aluminum tubing, with frangible disc type or resettable spring type pressure relief devices in each end. Each cylinder must conform to the Welker, Inc. sketches, photographs, drawings, and calculations submitted with the application and on file with the Office of Hazardous Materials Safety (OHMS). Each cylinder must conform with the DOT Specification 3E (§§ 178.35 and 178.42), except as follows:

§ 178.35(e) *Safety devices*.

Add: Each head must be equipped with a pressure relief device.

§ 178.35(c) *Duties of inspector*.

(1) * * *

(2) Add:

(iv) Verify chemical analysis of each heat of aluminum by analysis and check analysis of samples taken from one cylinder out of each lot of 200 or less completed cylinders.

(3) and (4) * * *

§ 178.35(g) *Inspector's report.*

Add: The inspector's report may be revised to accommodate the tests required by this special permit. A copy of the inspector's report on the first lot out of each design must be submitted to the OHMS.

§ 178.42(a) *Type, size and service pressure.* Seamless cylinder conforming with the following:

(1) Model CP-35 piston style receptacle has flat head on each end restrained with tie rods secured at each end by nuts or with tie rods threaded into an end cap and secured by nuts on the other end; maximum volume may not exceed 1,000 cc.

(2) Dimensions are as follows:

Model Nos.	Maximum O.D. x Length	Minimum Side Wall Thickness
CP-35	2.0" x 29"	0.11"

(3) Service pressure must be 600 pounds per square inch.

§ 178.42(b) *Aluminum.* All cylinder components must be 6061-T6 aluminum with a minimum yield strength of 40,000 psi and minimum ultimate strength of 45,000 psi.

§ 178.42(d) *Manufacture.*

Add: Welding, brazing, or heating to temperatures above 300 °F for any purpose whatsoever is prohibited.

§ 178.42(f) *Hydrostatic test.* Hydrostatic test must be as follows:

(1) Each cylinder (seamless tube with end caps but without pistons, and with tie bolts, if applicable) must be tested to a hydrostatic test of 1,200 psi and show no defect.

(2) In addition, one cylinder (seamless tube with end caps but without pistons, and with tie bolts, if applicable) out of each lot of 500 cylinders or less must be pressurized to failure. The pressure at which the cylinder fails to hold test pressure shall be recorded. This pressure may not be less than 2,400 psig. Rate of pressurization may not exceed 75 psi per second.

(3) The tube body of one cylinder from each lot that has passed the tests prescribed in § 178.42(f)(a)(2) of this special permit, must be flattened to not greater than 10 times wall thickness without cracking.

§ 178.42(g) *Leakage test*. Each compartment of each piston style receptacle within the fully assembled cylinder must be subjected to a pneumatic test of at least 200 psig without leakage.

§ 178.42(i) *Marking*. Applies except that “DOT-SP 10370” be stamped in lieu of “DOT 3E 1800”.

b. TESTING: Each cylinder must be visually reinspected at least once a year for deterioration of seals, scratches, dents, and gouges and must be cleaned as needed with an appropriate solvent recommended by the manufacturer. Components that are deteriorated or damaged shall be replaced by factory specified parts as shown on Welker, Inc. drawings and IOM manuals. Components may be replaced by the factory or the user, observing Tie Rod Nut tightening torque values shown in the Welker, Inc. IOM manuals. Any additional inspections or retesting requirements should follow the guidance of §§ 180.205 and 180.209.

8. SPECIAL PROVISIONS:

a. In accordance with the provisions of Paragraph (b) of § 173.22a, persons may use the packaging authorized by this special permit for the transportation of the hazardous materials specified in paragraph 6, only in conformance with the terms of this special permit.

b. A person who is not a holder of this special permit, but receives a package covered by this special permit, may reoffer it for transportation provided no modification or change is made to the package and it is offered for transportation in conformance with this special permit, the HMR, and the ICAO TI.

c. A current copy of this special permit must be maintained at each facility where the package is offered or reoffered for transportation.

d. Each packaging manufactured under the authority of this special permit must be marked with a registration symbol designated by the Office of Hazardous Materials Safety for a specific manufacturing facility.

- e. A current copy of this special permit must be maintained at each facility where the package is manufactured under this special permit. It must be made available to a DOT representative upon request.
 - f. Each cylinder must be overpacked for shipment in accordance with § 173.301(a)(9).
 - g. The cylinder may be used only for oil-well and LP gas sampling service.
 - h. Once a cylinder is exposed to fire for any period of time, it is no longer covered by this special permit and may not be used for the transportation of hazardous materials.
 - i. Transportation of Division 2.1 hazardous materials (flammable gases) are not authorized aboard aircraft unless specifically authorized in the Hazardous Materials Table (§ 172.101).
 - j. All internal coatings used for the cylinders shall be compatible with both the hazardous material and the cylinder.
9. MODES OF TRANSPORTATION AUTHORIZED: Motor vehicle and cargo-only aircraft.
10. MODAL REQUIREMENTS: A current copy of this special permit must be carried aboard each aircraft or motor vehicle used to transport packages covered by this special permit. The shipper must furnish a current copy of this special permit to the air carrier before or at the time the shipment is tendered.
11. COMPLIANCE: Failure by a person to comply with any of the following may result in suspension or revocation of this special permit and penalties prescribed by the Federal hazardous materials transportation law, 49 U.S.C. 5101 et seq:
- o All terms and conditions prescribed in this special permit and the Hazardous Materials Regulations, Parts 171-180.
 - o Persons operating under the terms of this special permit must comply with the security plan requirement in Subpart I of Part 172 of the HMR, when applicable.
 - o Registration required by § 107.601 et seq., when applicable.

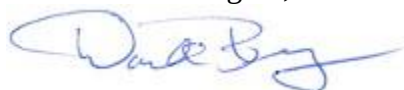
Each “Hazmat employee”, as defined in § 171.8, who performs a function subject to this special permit must receive training on the requirements and conditions of this special permit in addition to the training required by §§ 172.700 through 172.704.

No person may use or apply this special permit, including display of its number, when the special permit has expired or is otherwise no longer in effect.

Under Title VII of the Safe, Accountable, Flexible, Efficient Transportation Equity Act: A Legacy for Users (SAFETEA-LU) – “The Hazardous Materials Safety and Security Reauthorization Act of 2005” (Pub. L. 109-59), 119 Stat. 1144 (August 10, 2005), amended the Federal hazardous materials transportation law by changing the term “exemption” to “special permit” and authorizes a special permit to be granted up to two years for new special permits and up to four years for renewals.

12. **REPORTING REQUIREMENTS:** Shipments or operations conducted under this special permit are subject to the Hazardous Materials Incident Reporting requirements specified in 49 CFR §§ 171.15 - Immediate notice of certain hazardous materials incidents, and 171.16 - Detailed hazardous materials incident reports. In addition, the grantee(s) of this special permit must notify the Associate Administrator for Hazardous Materials Safety, in writing, of any incident involving a package, shipment or operation conducted under terms of this special permit.

Issued in Washington, D.C.:



for William Schoonover
Associate Administrator for Hazardous Materials Safety

Address all inquiries to: Associate Administrator for Hazardous Materials Safety, Pipeline and Hazardous Materials Safety Administration, Department of Transportation, Washington, D.C. 20590. Attention: PHH-13.

Copies of this special permit may be obtained by accessing the Hazardous Materials Safety Homepage at <https://www.phmsa.dot.gov/approvals-and-permits/hazmat/special-permits-search>. Photo reproductions and legible reductions of this special permit are permitted. Any alteration of this special permit is prohibited.

PO: AC/NICKS